

BDAT 614: Statistical Quality Control and Improvement

Sample Questions

Prof. Louis Asiedu

June 18, 2026

Detailed Calculation Questions and Conceptual Problems

Section A: Fundamentals of Quality and TQM (Conceptual)

1. Define “Quality” according to Joseph Juran, Philip Crosby, and W. Edwards Deming. Discuss how these definitions influence modern quality management practices.
2. Explain David Garvin’s **eight dimensions of quality**. Using a manufacturing product of your choice, illustrate how different stakeholders may prioritize different dimensions.
3. Differentiate clearly between **Quality Control (QC)**, **Quality Assurance (QA)**, and **Statistical Quality Control (SQC)**. Provide practical examples from a production environment.
4. Describe the key principles of **Total Quality Management (TQM)** and explain how Deming’s PDCA cycle and 14 Points support the implementation of TQM in an organization.
5. What is ISO 9001? Explain its purpose, key principles, and benefits of certification for a manufacturing company operating in a Big Data environment.

Section B: Process Variation and SPC

6. Distinguish between **Common Cause Variation** and **Special Cause Variation**. Why is it critical to distinguish between them before attempting process improvement or capability analysis?
7. Explain the **6Ms (Fishbone)** as sources of variation. Apply the 5 Whys technique to investigate a real manufacturing problem (e.g., frequent machine breakdown).
8. Describe the objectives and benefits of **Statistical Process Control (SPC)**. How does SPC support continuous improvement in a Big Data-driven manufacturing environment?

Section C: Control Charts

9. Explain the components of a control chart (Center Line, UCL, LCL). List and explain **four** Western Electric or Nelson rules for detecting out-of-control signals.
10. Compare $\bar{X}$ -**R Chart** and $\bar{X}$ -**S Chart**. When would you prefer one over the other? Include the relevant constants (A_2 , D_3 , D_4 , A_3 , B_3 , B_4) in your explanation.
11. Describe the **I-MR Chart** and explain why it is particularly useful in Big Data and sensor-based applications. Include the relevant constants (E_2 and D_4).
12. Differentiate between **p-chart**, **np-chart**, **c-chart**, and **u-chart**. For each, state the data type, sample size requirement, and a practical application.

Section D: Measurement System Analysis & Process Capability

13. Explain **Repeatability** and **Reproducibility**. Why is a Gauge R&R study a prerequisite for Process Capability Analysis?
14. In a Gauge R&R study, the results showed **%GR&R = 8.18%** and **ndc = 17.17**. Interpret these results and recommend actions to management.
15. Define **Cp**, **Cpk**, **Pp**, and **Ppk**. Using formulas, explain the difference between short-term and long-term capability. What does a $Cpk = 0.85$ indicate?
16. A process has $LSL = 45$, $USL = 55$, process mean = 51.2, short-term $\sigma = 1.8$, and long-term $\sigma = 2.3$. Calculate Cp , Cpk , Pp , and Ppk . Interpret the results and recommend next steps.

Section E: Acceptance Sampling

17. Define the following terms in Acceptance Sampling: AQL, LTPD, Producer's Risk (α), Consumer's Risk (β), and Operating Characteristic (OC) Curve.
18. Compare **Single**, **Double**, **Multiple**, and **Sequential** sampling plans in terms of procedure, Average Sample Number (ASN), complexity, and discrimination power.
19. Draw and interpret an OC Curve. Explain how sample size affects the steepness of the curve and the risks involved.

Section F: Integrated Case Studies (Application)

Case Study 1 (Variation + SPC + Capability + Gauge R&R)

A beverage company receives complaints about inconsistent bottle fill volumes. A Gauge R&R study gave $\%GR\&R = 12\%$ and $ndc = 8$. Control charts show the process is stable with $\bar{X} = 500.8$ ml and short-term $\sigma = 2.1$ ml. Specification limits are 495–505 ml.

1. Calculate Cp and Cpk .
2. Interpret the Gauge R&R and capability results.
3. Recommend a comprehensive improvement plan using concepts from SPC, 5 Whys, and DMAIC.

Case Study 2 (TQM + 5 Whys + Acceptance Sampling)

A manufacturing plant experiences high rejection rates of incoming raw materials. Management wants to reduce external failure costs.

1. Apply TQM principles to address this issue.
2. Use the 5 Whys technique to explore possible root causes.
3. Design a suitable acceptance sampling plan (Single or Double) with justification, including AQL and LTPD considerations.

Case Study 3 (Control Charts + Capability + Big Data)

A smart factory collects real-time sensor data from 500 machines. The process is stable but $Cpk = 0.95$.

1. Recommend appropriate control charts for continuous sensor data.
2. Suggest how Big Data tools (Kafka, Spark) can be used to enhance SPC and capability monitoring.
3. Propose a DMAIC project to improve the process.

Subgroup	$\bar{X}$ (ml)	R (ml)
1	501.2	4.1
2	498.8	3.5
3	500.5	5.2
4	502.1	6.8
5	499.7	3.9
6	501.8	4.7
7	500.3	2.8
8	503.4	7.1
9	499.5	4.3
10	501.0	5.0
11	500.8	3.6
12	502.6	6.2

Table 1: Bottling Process Data

Calculation Questions

Question 1: $\bar{X}$ and R Control Chart

A bottling company fills bottles with a target volume of 500 ml. The following data were collected from 12 subgroups of size $n = 5$:

Required:

- Calculate $\bar{\bar{X}}$ and $\bar{R}$.
- Using constants for $n = 5$ ($A_2 = 0.577$, $D_3 = 0$, $D_4 = 2.114$), compute UCL and LCL for $\bar{X}$ and R charts.
- Determine if the process is in statistical control.

Question 2: p-Chart (Attribute Data)

The following data were collected from LED bulb inspection:

Batch	n_i	D_i
1	200	8
2	250	12
3	180	5
4	220	15
5	300	9
6	190	14
7	240	7
8	210	11

Table 2: LED Bulb Inspection Data

Required:

- Calculate $\bar{p}$.
- Compute UCL and LCL for each batch.
- Determine if the process is in control and interpret.

Question 3a: Gauge R&R using Average & Range Method

A company performed a Gauge R&R study on shaft diameter with:

- 10 Parts, 3 Operators, 2 Replicates
- Grand Average = 50.12 mm
- $\bar{R} = 0.068$ mm

- Operator Average Difference (X_{diff}) = 0.13 mm

Constants: $d_2 = 1.128$, $K_2 = 0.5231$, $K_3 = 0.3146$

Required: Calculate EV, AV, Gauge R&R, %GR&R, and ndc. Interpret the results.

Question 3b: Gauge R&R using ANOVA Method

The ANOVA table from the same Gauge R&R study is as follows:

Source	DF	SS	MS	F
Parts	9	12.845	1.427	285.4
Operators	2	0.285	0.1425	28.5
Part $\times$ Operator	18	0.095	0.00528	1.056
Repeatability	60	0.300	0.005	—
Total	89	13.525	—	—

Table 3: Gauge R&R ANOVA Table

Required:

- Calculate all variance components (σ^2).
- Compute standard deviations.
- Calculate % Repeatability, % Reproducibility, and % Gauge R&R.
- Calculate ndc.
- Interpret the results and give recommendations.

Question 4: Process Capability Analysis

A steel rod process has:

- LSL = 49.80 mm, USL = 50.20 mm
- $\mu = 50.05$ mm
- Short-term $\sigma = 0.04$ mm
- Long-term $\sigma = 0.055$ mm

Required: Calculate C_p , C_{pk} , P_p , and P_{pk} . Interpret all indices and recommend actions.

Question 5: Integrated Case Study

A pharmaceutical company produces tablets with specification 500 mg $\pm$ 10 mg. Gauge R&R gave %GR&R = 18% and ndc = 6. The process is stable with $\bar{X} = 502$ mg and short-term $\sigma = 3.5$ mg. A lot of 2000 tablets was inspected using single sampling (n=125, c=4).

Required:

- Calculate Cp and Cpk.
- Interpret Gauge R&R results.
- Explain the single sampling plan decision rule and associated risks.
- Propose a DMAIC project incorporating at least three course concepts.

Question 6: c-Chart

The number of surface defects on 12 metal sheets: 4, 7, 3, 8, 5, 6, 2, 9, 4, 5, 7, 6

Required: Calculate $\bar{c}$, UCL, LCL, and interpret whether the process is in control.

Question 7: Gauge R&R Analysis – Comparison of Average & Range vs. ANOVA Method (with Raw Data)

A manufacturing company performed a crossed Gauge R&R study on the measurement system for machined shaft diameters. The study design is as follows:

- **10 Parts** (selected to represent the full process variation)
- **3 Operators** (A, B, C)
- **2 Replicates** per operator per part
- Total measurements = 60

The raw data (in mm) are provided below:

Table 4: Raw Gauge R&R Data

Part	Operator A		Operator B		Operator C	
	Rep 1	Rep 2	Rep 1	Rep 2	Rep 1	Rep 2
1	50.12	50.08	50.15	50.10	50.09	50.11
2	49.85	49.82	49.90	49.87	49.80	49.83
3	50.45	50.48	50.50	50.47	50.40	50.43
4	49.95	50.00	50.05	49.98	49.92	49.97
5	50.25	50.22	50.30	50.27	50.20	50.23
6	50.05	50.08	50.12	50.09	50.02	50.06
7	49.75	49.78	49.82	49.80	49.70	49.73
8	50.35	50.32	50.40	50.38	50.30	50.33
9	50.15	50.18	50.20	50.17	50.12	50.14
10	49.65	49.68	49.72	49.70	49.60	49.63

Required:

- Using the **Average & Range Method**, calculate EV, AV, % Gauge R&R, and ndc (Use constants: $d_2 = 1.128$, $K_2 = 0.5231$, $K_3 = 0.3146$).
- Using the **ANOVA Method**, construct the ANOVA table (or use appropriate software/formulas) and calculate the variance components: $\sigma^2_{\text{Repeatability}}$, $\sigma^2_{\text{Reproducibility}}$, $\sigma^2_{\text{Part-to-Part}}$, % Repeatability, % Reproducibility, % Gauge R&R, and ndc.
- Compare the results from both methods. Discuss differences, advantages, and limitations.
- Assess acceptability according to industry standards.
- Recommend the preferred method for Big Data / Industry 4.0 environments.

Questions on Acceptance Sampling

Question 8: Acceptance Sampling Fundamentals & OC Curve

- Define AQL, LTPD, Producer's Risk (α), and Consumer's Risk (β).
- Explain the Operating Characteristic (OC) Curve. What does the steepness indicate?
- Sketch a conceptual OC Curve and label AQL with $1 - \alpha$ and LTPD with β .

Question 9: Single Sampling Plan (Calculation)

A company receives lots of 2000 electronic components using $n = 125$, $c = 4$, with $p = 0.02$.

- State the decision rule.
- Calculate P_a using Poisson approximation.
- Estimate α and β if AQL=1% and LTPD=5%.

- d. Interpret protection for producer and consumer.

Question 10: Double Sampling Plan

Parameters: $n_1 = 80$, $c_1 = 2$, $r_1 = 5$; $n_2 = 80$, $c_2 = 6$.

- Explain the step-by-step decision procedure.
- Compare advantages/disadvantages vs. single sampling.
- Discuss ASN advantage.

Question 11: Comparison of Sampling Plans

Fill the comparison table (High/Medium/Low) for Single, Double, Multiple, and Sequential plans regarding ASN, complexity, discrimination power, and inspection cost. Justify one plan for Big Data real-time inspection.

Question 12: Integrated Acceptance Sampling Application

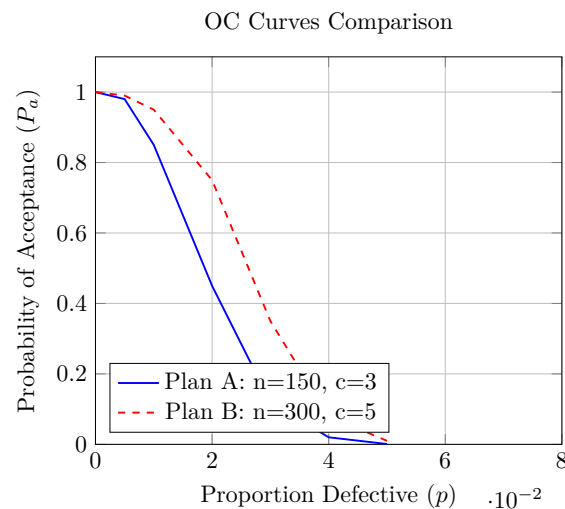
A pharmaceutical company wants a plan with AQL=0.5%, LTPD=3%, $\alpha \leq 5\%$, $\beta \leq 10\%$.

- Recommend Single or Double plan and justify.
- Discuss integration with Gauge R&R, Capability, and Control Charts.
- Explain how Big Data tools improve acceptance sampling.

Question 13: Operating Characteristic (OC) Curve Interpretation

Two single sampling plans for lots of 5000 units:

- **Plan A:** $n = 150$, $c = 3$
- **Plan B:** $n = 300$, $c = 5$



Required:

- Which plan has the steeper OC curve? What does it indicate about discriminatory power?
- Approximate Producer's Risk (α) at $p = 1\%$ (AQL) for both plans.
- Approximate Consumer's Risk (β) at $p = 4\%$ for both plans.
- Which plan better protects the consumer? Which better protects the producer?
- In a Big Data environment with automated inspection, discuss when acceptance sampling remains useful.